

Customer Shopping Behavior Analysis

Business Analysis Report | Python | MySQL | Power BI

1. Executive Summary

This project analyzes customer shopping behavior to understand shopping patterns, revenue drivers, customer engagement, discount usage and demographic contribution. The analysis starts with exploratory data analysis and data preparation in Python, moves the cleaned dataset into MySQL for structured business queries, and presents the resulting KPIs and comparisons in a Power BI dashboard.

2. Business Problem

The business needs a clear view of who its customers are, what they purchase, how much they spend, which categories and products contribute most to revenue, how discounts and subscriptions relate to spending, and which customer groups deserve greater attention. The goal is to turn a raw customer-shopping dataset into actionable evidence for marketing and customer-retention.

3. Business Objectives

- Measure overall customer base, average purchase value, and average review rating.
- Compare revenue contribution across gender, categories, age groups, discount status, and subscription status.
- Identify high-performing products and product-level opportunities.
- Evaluate shipping preferences, discount usage, etc.
- Build an interactive Power BI dashboard so decision-makers can filter the analysis by key customer dimensions.

4. Dataset Overview

Metric	Value
Records	3,900
Original columns	18
Age range	18–70
Average age	44.07 years
Average purchase amount	\$59.76
Average review rating	3.75 / 5
Unique categories	4
Unique products	25
Unique locations	50
Seasons	4
Shipping types	6
Payment methods	6
Purchase frequencies	7

The notebook has 18 original columns including customer demographics, product/category attributes, purchase amount, season, review rating, subscription, shipping, discount, previous purchases, payment method, and purchase frequency.

5. End-to-End Methodology

Stage	Work completed
1. Python / Pandas	Loaded the dataset, inspected structure, data types and missing values.
2. Data cleaning	Filled missing Review Rating values using the median rating within each category; standardized column names.
3. Feature engineering	Created age_group using quartiles and purchase_frequency_days from purchase-frequency labels.
4. Redundancy check	Confirmed Promo Code Used matched Discount Applied and removed the redundant field.
5. MySQL	Loaded the cleaned dataframe into the customer table in the customer_behaviour database.
6. SQL analysis	Answered 10 business questions covering revenue, products, shipping, subscriptions, discounts, segmentation, and age groups.
7. Power BI	Built KPI cards, charts, and slicers for interactive business reporting.

6. Data Cleaning and Feature Engineering

Area	Finding	Treatment
Missing Review Rating	37 values were missing.	Imputed using the median Review Rating within each Category.
Promo Code Used	It matched Discount Applied for all checked records.	Removed as a redundant column.
Column naming	Original names contained spaces and a USD-specific purchase field name.	Converted to lowercase, replaced spaces with underscores, and renamed purchase_amount_(usd) to purchase_amount.
Age grouping	No predefined age-group field existed.	Created quartile-based groups: Young Adult, Adult, Middle-aged, Senior.

The notebook confirms that all missing Review Rating values were resolved after the category-level median imputation. It also confirms that Promo Code Used and Discount Applied were identical for all checked records, so the redundant field was removed.

7. SQL Analysis Framework

The SQL script contains 10 business questions. These move beyond simple totals and cover customer comparison, product performance, shipping, subscription behavior, discount intensity, customer segmentation, category-level product ranking, and age-group revenue.

Business Questions:

7.1 Male vs. female revenue

gender	revenue
Male	157890
Female	75191

7.2 Customers using discounts who spent above average

customer_id	purchase_amount
2	64
3	73
4	90
7	85
9	97
12	68
13	72
16	81
20	90
22	62

7.3 Top 5 products by average review rating

item_purchased	Average Product Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.8
Skirt	3.78

7.4 Average purchase amount: Standard vs. Express shipping

shipping_type	purchase_amount
Express	60.48
Standard	58.46

7.5 Subscriber vs. non-subscriber spend and revenue

subscription_status	count(customer_id)	round(avg(purchase_amount),2)	sum(purchase_amount)
Yes	1053	59.49	62645
No	2847	59.87	170436

7.6 Products with the highest percentage of discounted purchases

item_purchased	discount_rate
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

7.7 New / Returning / Loyal customer segmentation

customer_segment	Number of Customers
Loyal	3116
Returning	701
New	83

7.8 Top 3 most purchased products within each category

item_rank	category	item_purchased	total_orders
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

7.9 Relationship between repeat buying and subscription status

subscription_status	repeat_buyers
Yes	958
No	2518

7.10 Revenue contribution by age group

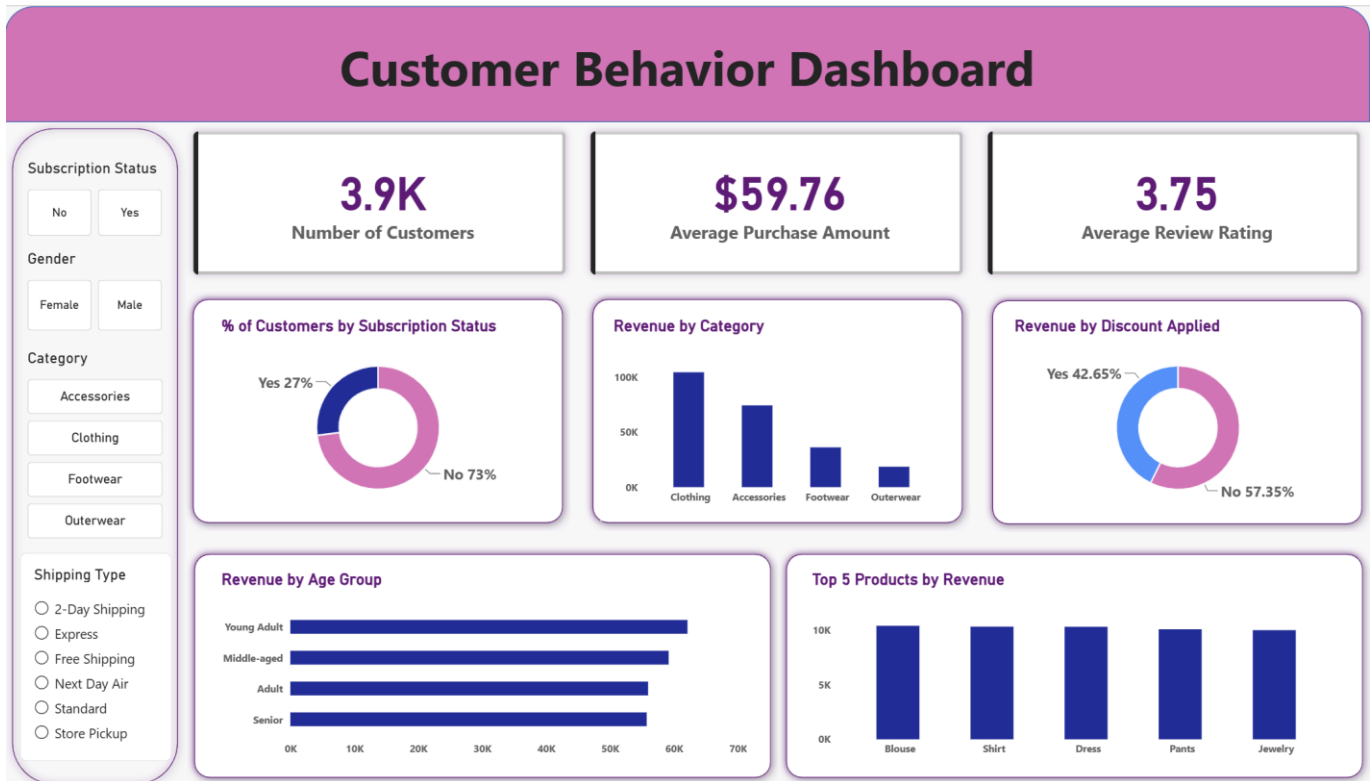
age_group	total_revenue
Young Adult	62143
Middle-aged	59197
Adult	55978
Senior	55763

8. Power BI Dashboard

The supplied Power BI page combines KPI cards, comparative visuals, and slicers. The dashboard is designed as a single-page management view with filters for Subscription Status, Gender, Category, and Shipping Type.

Element	Dashboard component	Observed result / purpose
KPI	Number of Customers	3.9K
KPI	Average Purchase Amount	\$59.76
KPI	Average Review Rating	3.75
Visual	% of Customers by Subscription Status	27% Yes / 73% No
Visual	Revenue by Category	Clothing is the leading category

Visual	Revenue by Discount Applied	42.65% discounted / 57.35% non-discounted purchase amount
Visual	Revenue by Age Group	Young Adult leads at about \$62K
Visual	Top 5 Products by Revenue	Product-level prioritization
Slicers	Subscription, Gender, Category, Shipping Type	Interactive filtering



9. Key Business Insights

Customer base and spending

The dataset contains 3,900 customer records. The average purchase amount is \$59.76, while the average review rating is 3.75/5. This gives the business a useful baseline for monitoring customer value and satisfaction.

Subscription penetration

The dashboard shows 27% of customers as subscribers and 73% as non-subscribers. The non-subscriber majority represents a substantial pool for retention and subscription-conversion analysis.

Category performance

Clothing contributes the highest revenue among the four categories, followed by Accessories, with Footwear and Outerwear contributing less. Category-level inventory, merchandising, and campaign planning should therefore give particular attention to the leading categories.

Discount behavior

42.65% of purchase amount is associated with transactions where a discount was applied, compared with 57.35% without a discount. This describes revenue mix by discount status; it does not measure discount profitability because discount value, cost, and margin are not available.

Age-group contribution

Young Adults contribute the highest revenue at approximately \$62K, followed by Middle-aged customers at about \$59K, with Adult and Senior groups at about \$56K each. Age is therefore useful for campaign segmentation, while the differences should be interpreted as descriptive rather than causal.

Customer loyalty

The SQL analysis explicitly segments customers as New, Returning, and Loyal using previous purchases and separately examines repeat buyers and subscription status. This creates a foundation for retention and loyalty strategies.

10. Recommendations

1. Increase subscription conversion

Focus conversion campaigns on the 73% non-subscriber base, especially customer groups that show stronger repeat-purchase behavior. Before launching a broad incentive, use the existing subscription and repeat-buyer SQL analysis to identify the highest-potential segments.

2. Optimize discounts instead of simply increasing them

Because a substantial share of purchase amount occurs with discounts, evaluate discount effectiveness at product level. Prioritize products with high discount rates and compare their average purchase values before expanding promotions. Profit-margin data would be needed to determine true ROI.

3. Protect and expand leading categories

Clothing is the strongest revenue category, so inventory availability, merchandising, cross-selling, and campaign visibility should be monitored closely for this category. Accessories can be used as a complementary growth area.

4. Use age groups for targeted marketing

Young Adults are the highest-revenue age group in the dashboard. Create targeted campaigns for this segment while maintaining separate messaging for Adult, Middle-aged, and Senior customers.

5. Evaluate shipping decisions with cost data

The SQL analysis compares Standard and Express average purchase amounts. A commercial decision about shipping should additionally consider shipping cost, delivery performance, and customer satisfaction because those measures are not present in the dataset.

11. Technical Stack

Tool	Role
Python / Pandas	Data loading, EDA, cleaning, missing-value treatment, feature engineering
MySQL	Storage of the cleaned customer table
MySQL	Business-question analysis and customer/product segmentation
Power BI	Interactive KPI dashboard, filtering, and business presentation